

Glengarry Pl lower deck

· 2'4" high in the alcove ·
parapet walls stay, no rail

166 Glengarry Pl, Castle Rock · Vintage Coastline, Dark Hickory picture frame ·
SYP @ 12" · 6x12 DF beam · Cortex

CHECK OR ACH	FINANCED	MONTHLY	SQUARE FEET	FOOTINGS	RAIL
\$37,866	\$40,716	\$473	332.8	3	0 LF

DESIGN READS

Frame	2 zones, 19'-1" along the house, 18'-4" at the deepest
Deck	29.2 LF of outer edge · 332.8 SF · 2'-4" above grade
Zone	S Step — 2'1" beside the alcove's west wall, 10'2" deep to the front: 2'-1" x 10'-2"
Zone	A Alcove — 17' wall, 18'4" deep, ends at the east wing wall: 17'-0" x 18'-4"
Joists	21 2x10 #1 True Frame Joist SYP GC @ 12" OC
Rims	2-ply 2x10 front and sides
Beam	DROP 6x12 DF S+A: 19'-0", 3 posts at 1'-0 1/2" / 9'-6 1/2" / 18'-0 1/2" (worst 4,343 lb)
Posts	3 x 6x6 on DP-50/50
Load	50 psf (40 psf live governs) · engineered
Decking	TimberTech Vintage Coastline — 38 rows parallel to the house, picture frame in Vintage Dark Hickory, 3/16" gaps, Cortex
Rail	Existing stucco parapet stays — no rail in this scope

- Rows per panel: panel 1 (0"-19'-1") 38 rows · 1 ripped pieces · no breakers
- DROP 6x12 DF S+A: 19'-0" long, 3 posts at 1'-0 1/2" / 9'-6 1/2" / 18'-0 1/2"
- S: 2'-2 3/4" cantilever past the drop beam exceeds L/4 of the 7'-9 3/4" back-span — engineered condition (owner-directed)
- A: post load 3,811 lb exceeds DP-50/50 (3,300 lb) — upgraded to DP-75/63

ENGINEER	ENGINEERING	DROP 6x12 DF: 6x12 not in IRC R507.5 — treated as (3)2x12; engineer to confirm; 6x12 not in IRC R507.5 — treated as (3)2x12; engineer to confirm IRC Table R507.5
ENGINEER	ENGINEERING	Cantilever 2'-2 3/4" > L/4 of the 7'-9 3/4" back-span — owner-directed, in the stamped set IRC R507.5 / R507.6
CODE	GUARD	existing stucco parapet reused as the guard — verify 36" min above the new deck surface and 200 lb top-rail load; if the new deck surface rises, the parapet height drops IRC R312.1
CODE	FIRE	WUI zone: Vintage is Ignition Resistant, Class A, WUI — keep the under-deck area clear of combustibles, 1-hr or noncombustible enclosure may be required IWUIC 504.7
CHECK	SETBACK	rear setback not entered — confirm the required rear setback and the deck-to-line distance before the frame is laid out zoning
CHECK	SETBACK	side setback not entered — confirm the required side setback and the deck-to-line distance before the frame is laid out zoning
CHECK	FOOTING	Diamond Pier: pins 50" — locate utilities (811) before driving; bearing 1,611 lb/post vs 3,300 lb allowable (manufacturer chart, confirm)
CHECK	LEDGER	verify the rim board behind the ledger is solid 2x/LVL/engineered rim (not I-joist web or a hollow band) before the ledger fasteners go in IRC R507.9.1.1
INFO	LEDGER	lateral: 4 x DTT1Z (750 lb each) into house floor framing IRC R507.9.2
INFO	WIND	basic wind speed 115 mph — standard connectors
INFO	GUARD	deck 2'-4" above grade — under 30", guard not required by IRC (client choice)

THE ORDER Decks & Docks · GSEXT1 · ship to the job

Lumber

NET	ORDER	UNIT	ITEM	WHY
7	8	ea	2x10x10 #1 True Frame Joist SYP GC	+1 cull (field joist · 2 PF joist · 2 left rim ply · 2 right rim ply)
4	5	ea	2x10x12 #1 True Frame Joist SYP GC	+1 cull (blocking, front rim ply, ledger · 3 blocking)
25	26	ea	2x10x20 #1 True Frame Joist SYP GC	+1 cull (16 field joist · 2 PF joist · 2 left rim ply · 2 right rim ply · 2 front rim ply · ledger)
1	1	ea	6x12x20 Douglas Fir #2	exact (drop beam: DROP 6x12 DF S+A)
1	1	ea	6x6x8 #2 GC	exact (yields 3 posts at ~6")

Footings

NET	ORDER	UNIT	ITEM	WHY
3	3	ea	Diamond Pier DP-50/50 (head + 4 x 50" pins, plugs, caps)	exact — confirm stock
3	3	ea	Simpson ABA66Z adjustable post base 6x6 on the pier bolt	exact

Hardware

NET	ORDER	UNIT	ITEM	WHY
42	42	ea	Simpson LUS28Z face-mount hanger 2x8/2x10	exact (21 joists x 2 ends)
8	8	ea	Simpson HUCQ210-2-SDS concealed-flange double 2x10 hanger (SDS screws included)	exact (2 side rims x 2 ends) — confirm stock
1	1	box	Camo 9ga HDG hanger nail 3" 5 lb (~350)	252 needed of ~350
1	1	box	Camo 9ga HDG hanger nail 1-1/2" 5 lb (~650)	418 needed of ~650
1	1	box	FastenMaster LedgerLOK 3-5/8" 50 ct	21 needed
8	8	ea	Simpson DTT1Z lateral tie kit w/ SDS screws	exact (2 near each end of every ledger, into house floor framing)
25	25	ea	Simpson H2.5AZ hurricane tie (joists to the drop beam, hidden)	exact (21 joists + 2 rims at each drop beam)
3	3	ea	Simpson BC6Z post cap 6x post / 6x beam	exact
1	1	box	Simpson SD10212R100 #10 x 2-1/2" connector screw 100 ct	66 needed (bases 36 · caps 30)
4	4	box	GRK RSS #10 x 3-1/8" structural screw 100 ct	316 needed (laminations 164 · blocking 152)

Flashing & waterproofing

NET	ORDER	UNIT	ITEM	WHY
1	1	roll	Grace Vycor Deck Protector 12" x 75' (wall membrane behind ledger + side rims)	27 LF of house wall with laps — nothing on a privacy wall
3	3	ea	Aluminum L-flashing 10' (over ledger + side rims)	house walls only
3	3	ea	26 ga galvanized wall flashing 12" x 10' (CWRC 404.3.2: noncombustible flashing at the deck/wall intersection, behind cladding)	WUI practice — house walls only
7	7	roll	Nichigo G-Tape 3035 2" x 65' black	405' needed of 455'
2	2	roll	Nichigo G-Tape 3035 4" x 65' black	94' needed of 130'

Decking

NET	ORDER	UNIT	ITEM	WHY
38	40	ea	TimberTech Vintage Coastline 1x6x20 Square Edge	+2 (38 x 20' · panel 1 38 rows · every row one piece)
1	1	ea	TimberTech Vintage Dark Hickory 1x6x16 Square Edge — borders	exact (border 0",8'-2" 10'-2 15/16")
1	2	ea	TimberTech Vintage Dark Hickory 1x6x20 Square Edge — borders	+1 (border 0",18'-4" 19'-1 15/16")

Fasteners

NET	ORDER	UNIT	ITEM	WHY
5	5	box	Cortex for TimberTech PVC, Coastline plugs (350 ct / 100 SF)	1532 screws (2 per board at every bearing point) · 1750 supplied
1	1	box	Cortex for TimberTech PVC, Dark Hickory plugs — borders + dividers	50 screws

Fascia

NET	ORDER	UNIT	ITEM	WHY
7	8	ea	TimberTech Vintage Fascia Coastline 0.5x11.75x12	+1 (left fascia (S) 10'-2 1/2" · right fascia (S) 10'-2 1/2" · front fascia (S) 2'-2" · left fascia (A) 12'-0" · left fascia (A) 6'-4 1/2" · right fascia (A) 12'-0" · right fascia (A) 6'-4 1/2" · front fascia (A) 12'-0" · front fascia (A) 5'-1")

Site

NET	ORDER	UNIT	ITEM	WHY
330	330	SF	Demo existing deck + haul-off	labor line — field verify the existing deck

FASTENER & CONNECTOR SCHEDULE

Ledger	2x10 ledger x 19'-1" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 21 fasteners
Joist tape	G-Tape 2" every joist, PF joist, blocking and the ledger top · 4" on both rims and the beam(s)
Field	Cortex face screws with color-matched plugs: 2 per board at every joist and at both ends of every run, color-matched plugs (1532 field + 50 border/divider)

▼ Cut list

MEMBER	STOCK	LENGTH	QTY	NOTE
Field joists (S)	2x10	9'-9 1/2"	1	@ 12" OC, crown up
Picture-frame joists (S)	2x10	9'-9 1/2"	2	centre 5 1/4" from each frame face
Side rim plies (S)	2x10	9'-9 1/2"	4	2-ply, laminated on the ground with RSS 2 rows @ 12" staggered
Front rim plies (S)	2x10	2'-1"	2	inner ply takes the hangers; outer ply laminated after
Ledger (S)	2x10	2'-1"	1	top at deck height less board thickness; membrane behind, flashing over
Blocking (S)	2x10	10 1/2"	4	2 rows at 7'-11 1/4", 4'-0 3/8" from the wall
Field joists (A)	2x10	17'-11 1/2"	16	@ 12" OC, crown up
Picture-frame joists (A)	2x10	17'-11 1/2"	2	centre 5 1/4" from each frame face
Side rim plies (A)	2x10	17'-11 1/2"	4	2-ply, laminated on the ground with RSS 2 rows @ 12" staggered

MEMBER	STOCK	LENGTH	QTY	NOTE
Front rim plies (A)	2x10	17'-0"	2	inner ply takes the hangers; outer ply laminated after
Ledger (A)	2x10	17'-0"	1	top at deck height less board thickness; membrane behind, flashing over
Blocking (A)	2x10	10 1/2"	34	2 rows at 16'-1 1/4", 8'-1 3/8" from the wall
Field boards panel 1	Vintage Coastline	17'-0"	16	one board wall to border, no joints
Field boards panel 1 (rip)	Vintage Coastline	17'-0"	1	one board wall to border, no joints; ripped to fit the border / the house
Field boards panel 1	Vintage Coastline	18'-8"	21	one board wall to border, no joints
DROP 6x12 DF S+A beam	6x12	19'-0"	1	y 16'-1 3/16"; posts at 1'-0 1/2" / 9'-6 1/2" / 18'-0 1/2"
Posts under DROP 6x12 DF S+A	6x6	6"	3	field-measure each
Border 0",8'-2"	Vintage Dark Hickory square	10'-2 15/16"	1	mitre at the corners; dividers on the doubled joist
Border 0",18'-4"	Vintage Dark Hickory square	19'-1 15/16"	1	mitre at the corners; dividers on the doubled joist
Left fascia (s)	fascia	10'-2 1/2"	1	flush under the board nose; full board at the front corners, short piece at the house
Right fascia (s)	fascia	10'-2 1/2"	1	flush under the board nose; full board at the front corners, short piece at the house
Front fascia (s)	fascia	2'-2"	1	flush under the board nose; full board at the front corners, short piece at the house
Left fascia (a)	fascia	12'-0"	1	flush under the board nose; full board at the front corners, short piece at the house
Left fascia (a)	fascia	6'-4 1/2"	1	flush under the board nose; full board at the front corners, short piece at the house
Right fascia (a)	fascia	12'-0"	1	flush under the board nose; full board at the front corners, short piece at the house
Right fascia (a)	fascia	6'-4 1/2"	1	flush under the board nose; full board at the front corners, short piece at the house
Front fascia (a)	fascia	12'-0"	1	flush under the board nose; full board at the front corners, short piece at the house
Front fascia (a)	fascia	5'-1"	1	flush under the board nose; full board at the front corners, short piece at the house

THE STACK internal

Materials (raw)	\$11,449.33
Materials × 1.15	\$13,166.73
Sales tax 7.90% on factored materials	\$1,040.17
Labor (rate card)	\$6,800.40

Work — carries the margin	\$21,007.30
General conditions at cost	\$1,331.20
Cost	\$22,338.50
Sell — check or ACH at 42.5% GM	\$37,866 · \$113.78/SF
Sell at the 40% floor	\$36,343
Financed	\$40,716 · \$473/mo at 6.99% × 10 yr
Gross profit	\$15,528
Engineering — outside the price, at cost	\$1,800–\$3,000

Labor

LINE	QTY	UNIT	RATE	EXT
Demolition — existing deck	330	SF	\$2.00	\$660.00
Diamond Pier install (replaces caisson)	3	pier	\$50.00	\$150.00
Frame	332.8	SF	\$10.00	\$3,328.00
Decking & fascia	332.8	SF	\$8.00	\$2,662.40

General conditions — at cost, no margin

ITEM	AMOUNT	WHY
General conditions	\$1,331	\$4/SF x 332.8 SF

What the price includes — sell values

ITEM	VALUE
Demolition & haul-off	\$1,148
Foundation & structure	\$14,900
Decking & fascia	\$20,487
Site & project services	\$1,331
Total, check or ACH	\$37,866
Total, financed	\$40,716

THE QUOTE homeowner-facing · facts only

QUOTE · PRE-ENGINEERING

GLENGARRY PL — LOWER DECK

2x10 SYP frame at 12", 6x12 beam, 6x6 posts — on 3 Diamond Pier footings.
TimberTech Vintage Coastline decking with a Dark Hickory border.

332.8 SF · 2 zones · **2'-4"** above grade, engineered · **30 psf** snow design
Prepared for the owner of 166 Glengarry Pl, Castle Rock

PRICE

Total investment \$40,716 — financed, no money down

- 12 months no payments, no down payment
- 18 months no payments, no down payment
- 6.99% for 10 years — about \$473/month

Check or ACH: \$37,866 (7% savings)

WHAT THE PRICE INCLUDES

ITEM	VALUE
Demolition & haul-off	\$1,148
Foundation & structure	\$14,900
Decking & fascia	\$20,487
Site & project services	\$1,331
Total, check or ACH	\$37,866
Total, financed	\$40,716

Engineering — \$1,800 to \$3,000. Stamped drawings by a Colorado engineer (30 psf snow). GSX provides the engineer and bills the engineering at cost, with no markup. You own the stamped set. It is separate from the deck price above.

INCLUDED — WHAT IT IS

ELEMENT SELECTION

Decking	TimberTech PVC Vintage, Coastline; picture frame and dividers in Vintage Dark Hickory. Square-shoulder boards, 3/16" gaps, matching fascia. Ignition Resistant, Class A, WUI; Limited Lifetime product / 50-yr fade & stain.
Frame	21 2x10 #1 True Frame Joist SYP GC @ 12" OC; 2-ply 2x10 front and sides; DROP 6x12 DF S+A: 19'-0", 3 posts at 1'-0 1/2" / 9'-6 1/2" / 18'-0 1/2" (worst 4,343 lb).
Foundation	DP-50/50.
Design load	30 psf snow, stamped by a Colorado engineer. Colorado Wildfire Resiliency Code practice: Class A decking, noncombustible rail, metal flashing at every wall.
Hardware	Simpson ZMAX galvanized connectors. Cortex fasteners with color-matched plugs.

PLAN

2 zones · 332.8 SF · 19'-1" along the house · 2'-4" above grade

- **S · Step — 2'1" beside the alcove's west wall, 10'2" deep to the front — 2'-1" x 10'-2".**
- **A · Alcove — 17' wall, 18'4" deep, ends at the east wing wall — 17'-0" x 18'-4", no rail on the wall side.**

Zones are named left to right standing in the yard facing the house.

SCHEDULE

1. Engineering & ordering (2-3 weeks). Stamped drawings, materials ordered, samples approved.
2. Day 1 — protection, utilities located (811), old deck out.
3. Day 2 — layout, 3 Diamond Piers driven, ledger set.
4. Days 3-4 — posts and beams set.
5. Days 5-7 — frame, flashing at every wall.
6. Days 8-9 — decking, borders first.
7. Day 10 — final clean; end of construction.

NEXT STEP

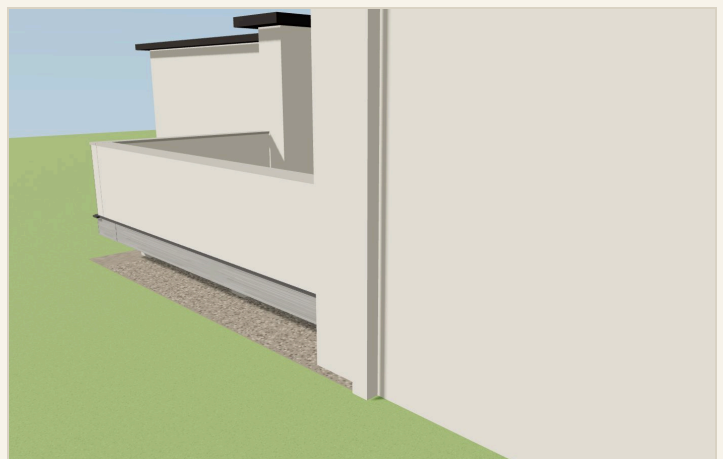
Engineering first. This is a quote: the numbers depend on the stamped engineering, so the first step is engineering. GSX provides the Colorado engineer and supplies the design package; you pay GSX the engineer's cost at no markup and you own the stamped set. When the stamped set is in hand, GSX issues a fixed-price proposal on these numbers with the payment terms.

Quote, not a contract: a fixed-price proposal follows the stamped engineering. Quote numbers hold 30 days. Renders and plan are design intent; the stamped engineering set governs structure, dimensions and locations.

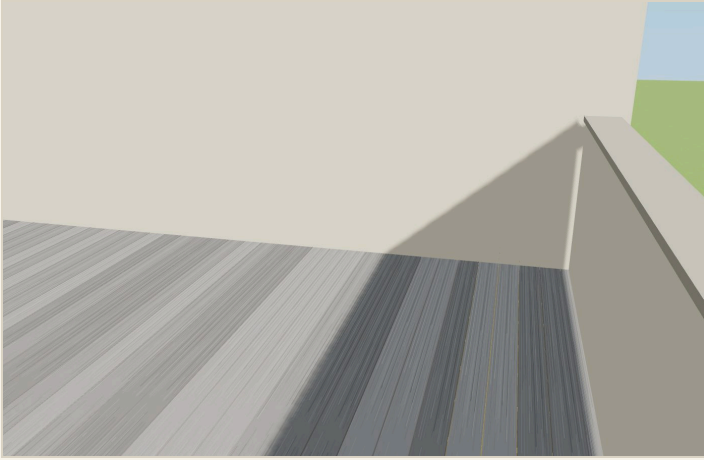
RENDERS 3D model · design intent · timbers shown as specified



From the yard



From the front corner

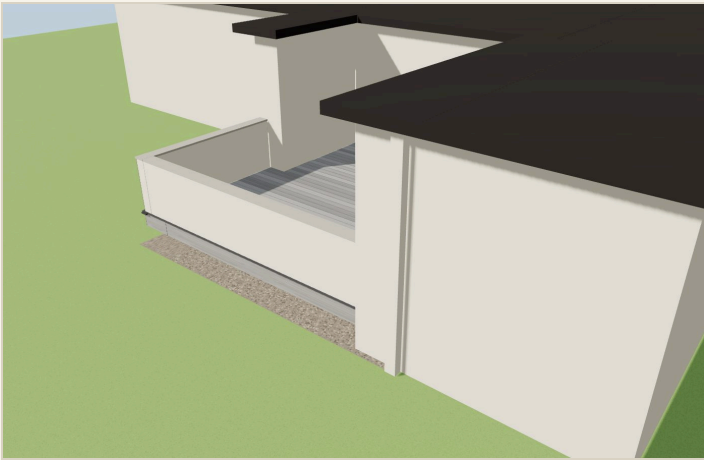


Standing on the deck



Under the deck — posts, beams, footings

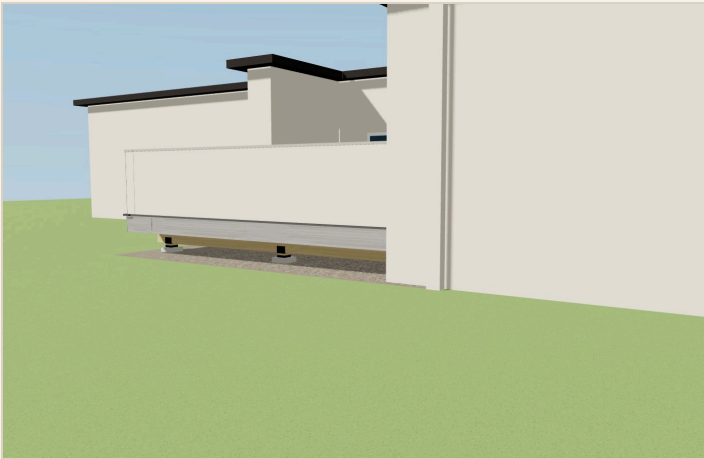
ISOMETRICS



Isometric from the stair corner



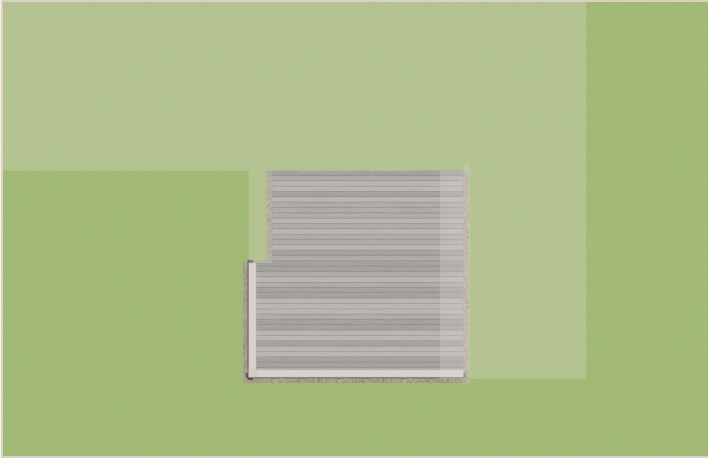
Isometric from the open corner



Low isometric

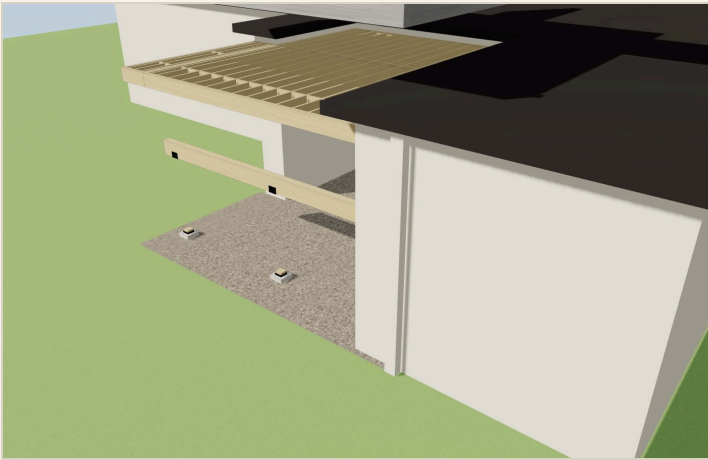


Isometric from under the frame

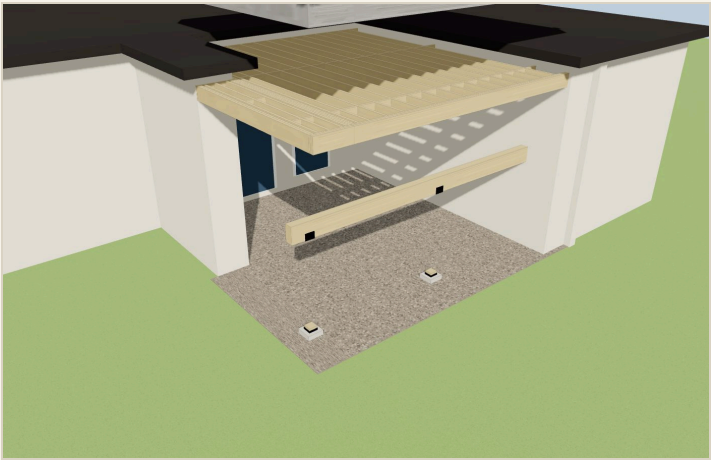


Plan from above

EXPLODED



Exploded — every layer



Exploded — from the open corner



Exploded — from the front corner



Exploded — low angle

G-001 General notes · design criteria

G-001 GENERAL NOTES · DESIGN CRITERIA

Loads, codes, structure summary, sheet index, site — the permit-set cover

Glengarry PI — lower deck · 166 Glengarry PI, Castle Rock

GSX · Jade Pender 303-550-9558 · September 17, 2026 · Rev 0 FOR REVIEW

PROJECT

Glengarry PI — lower deck
166 Glengarry PI, Castle Rock CO 80108
332.8 SF · 2'-4" above grade · 2 zones

DESIGN CRITERIA

Deck live load	40 psf uniform (IRC Table R301.5); guards 200 lb concentrated / 50 plf
Ground snow	30 psf (ground; deck design uses the larger of snow and live) — 40 psf live governs
Dead load	10 psf deck
Roof snow (cover)	—
Design total	50 psf on the deck
Wind	115 mph Vult, Exposure C, Risk Cat. II (ASCE 7-16) — confirm special wind region
Frost depth	3'-0" (footings bear below)
Soil bearing	1,500 psf presumptive (IRC R401.4.1) — no soils report
Seismic	B
Wildfire (WUI)	Ignition-resistant construction: Class A / WUI-listed decking (TimberTech Advanced PVC), noncombustible rail (steel), fiber-cement skirt, soffit and fascia, 26 ga flashing at every wall, Class A roof on the cover; under-deck kept clear of combustibles (Colorado Springs WUI 2021 IRC as adopted by Castle Rock; IRC R507 decks; ASCE 7-16 loads; IWUIC where mapped; Tables R507.5 / R507.6 / R507.4 / R507.9
Codes	Stamped structural set by a Colorado PE required — the stamped set governs
Engineering	

STRUCTURE

Footings	DP-50/50
Posts	3 x 6x6
Beams	DROP 6x12 DF S+A: 19'-0", 3 posts at 1'-0 1/2" / 9'-6 1/2" / 18'-0 1/2" (worst 4,343 lb)
Joists	21 2x10 #1 True Frame Joist SYP GC @ 12" OC
Rims	2-ply 2x10 front and sides
Ledger	2x10 ledger x 19'-1" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 21 fasteners
Decking	TimberTech Vintage Coastline — 38 rows parallel to the house, picture frame in Vintage Dark Hickory, 3/16" gaps, Cortex
Rail	Existing stucco parapet stays — no rail in this scope

SHEET INDEX

G-001	General notes · design criteria
A-001	Isometric · renders
S-100	Foundation plan
S-101	Framing plan
A-101	Decking plan
A-201	Railing plan + elevation
A-301	Section + front elevation
D-501	Details
T-400	Build steps (build set)
T-700	Order + schedules (takeoff)

SITE

Castle Rock (Douglas County) — combined sales tax 7.9%. Lower deck 2'4" above grade in the alcove under the upper deck; the stucco parapet walls stay, no rail. Ponderosa lot — WUI.

FLAGS (INTERNAL — remove before the permit submittal)

ENGINEER	DROP 6x12 DF: 6x12 not in IRC R507.5 — treated as (3)2x12;...
ENGINEER	cantilever 2'-2 3/4" > L/4 of the 7'-9 3/4" back-span — owner...
CODE	existing stucco parapet reused as the guard — verify 36" min...
CODE	WUI zone: Vintage is Ignition Resistant, Class A, WUI — keep the...

No scale

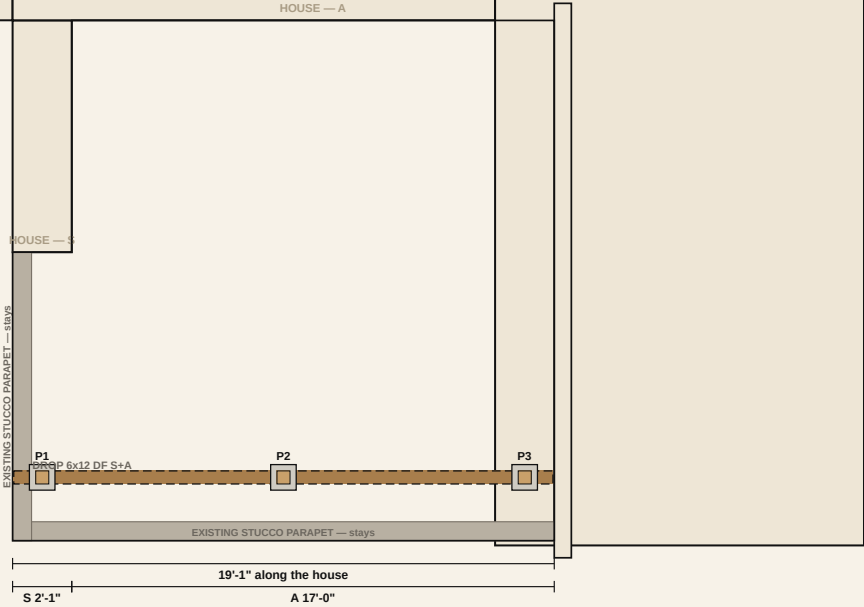
Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

FOOTING SCHEDULE

Diamond Pier DP-50/50 (head + 4 x 50" pins),
3,300 lb allowable

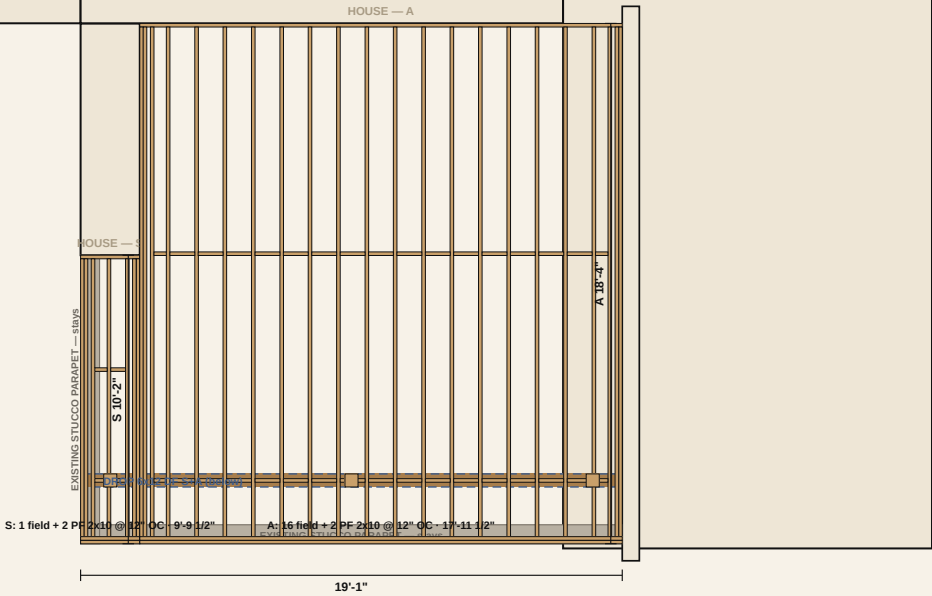
ID	X	Y	ZONE	LOAD
P1	1'-0 1/2"	16'-1 3/16"	S	1,611 lb
P2	9'-6 1/2"	16'-1 3/16"	A	4,343 lb
P3	18'-0 1/2"	16'-1 3/16"	A	2,682 lb

x from the left frame face facing the house; y from the reference wall.
Frost 3'-0" · soil 1,500 psf · worst post 4,343 lb



FRAMING NOTES

- 21 2x10 #1 True Frame Joist SYP GC @ 12" OC
- 2-ply 2x10 front and sides
- 2x10 ledger x 19'-1" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · Z1
- blocking one row over the beam, staggered
- G-Tape 2" every joist, PF joist, blocking and the ledger top · 4" on both rims and the beam(s)
- DROP 6x12 DF S+A: 19'-0", 3 posts at 1'-0 1/2" / 9'-6 1/2" / 18'-0 1/2" (worst 4,343 lb)
- Connectors:
 - Simpson LUS28Z face-mount hanger 2x8/2x10: 42
 - Simpson HUCQ210-2-SDS concealed-flange double 2x10 hanger (SDS screws included): 8
 - Simpson DTT11Z lateral tie kit w/ SDS screws: 8
 - Simpson H2.5AZ hurricane tie (joists to the drop beam, hidden): 25
 - Simpson BC6Z post cap 6x post / 6x beam: 3



A-101 Decking plan

A-101

DECKING PLAN

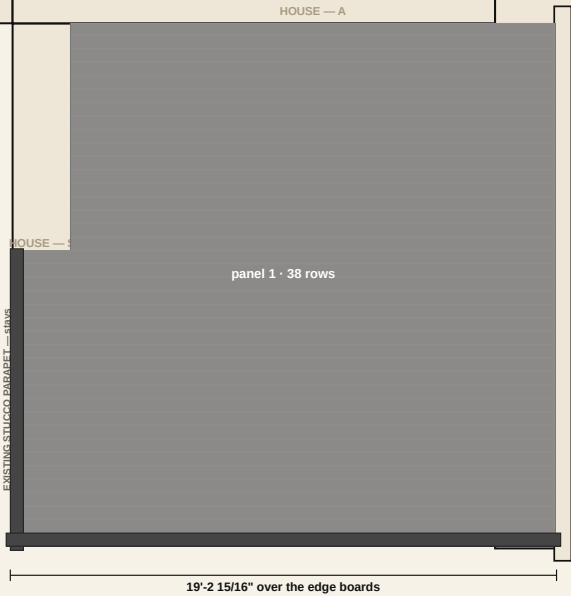
Every board, border and divider as it lays; rail posts shown for the pockets

Glengarry PI — lower deck · 166 Glengarry PI, Castle Rock

GSX · Jade Pender 303-550-9558 · September 17, 2026 · Rev 0 FOR REVIEW

DECKING

Field: TimberTech Vintage Coastline, 5.5" x 1", 3/16" gaps
Border: Vintage Dark Hickory
Boards run parallel to the house; edge overhang 1" over the fascia
Cortex face screws with color-matched plugs: 2 per board at every joist and at both ends of every run, color-matched plugs (1532 field + 50 border/divider)
Rows per panel: panel 1 (0"-19'-1") 38 rows · 1 ripped pieces · no breakers — every board one piece wall to border, no butt joints



Plan · not to scale · house at the top · dimensions govern

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

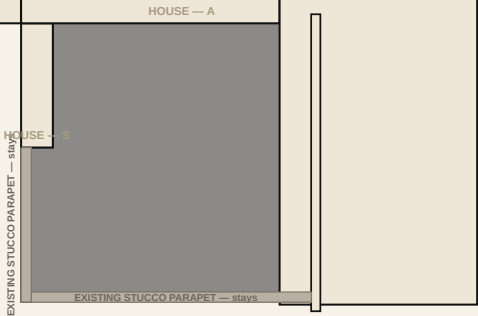
A-201 Railing plan + elevation

A-201 RAILING PLAN + ELEVATION

Post IDs and types (E end · L line · C corner), bay lengths, outer-edge elevation

Glengarry PI — lower deck · 166 Glengarry PI, Castle Rock
GSX · Jade Pender 303-550-9558 · September 17, 2026 · Rev 0 FOR REVIEW

RAILING
Existing stucco parapet stays — no rail
in this scope



Plan above · elevation of the outer edge below

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

A-301 Section + front elevation

A-301 SECTION + FRONT ELEVATION

Cut through the deepest zone; heights to grade, frost line, cantilever, guard

Glengarry PI — lower deck · 166 Glengarry PI, Castle Rock
GSX · Jade Pender 303-550-9558 · September 17, 2026 · Rev 0 FOR REVIEW

SECTION through zone A at 8'-6" — looking toward the left end

FRONT ELEVATION — from the yard

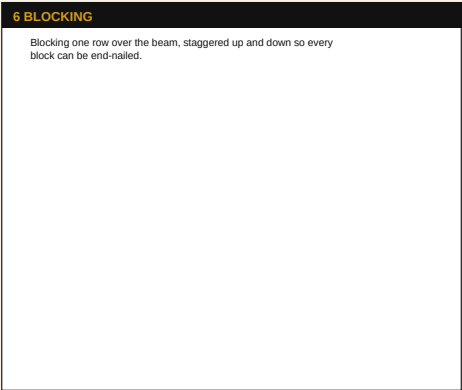
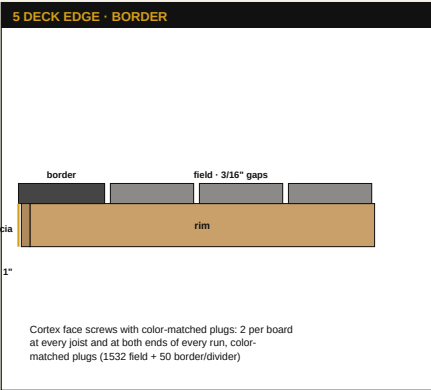
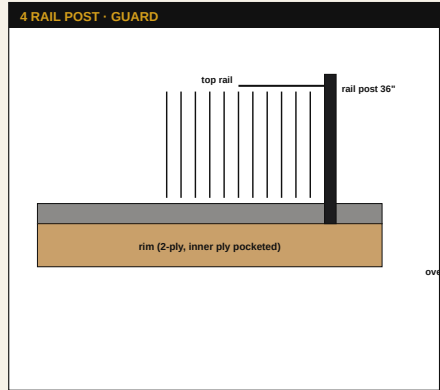
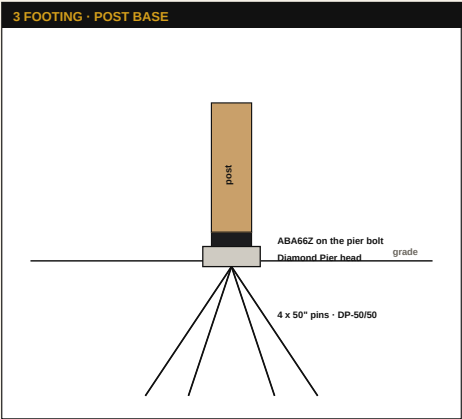
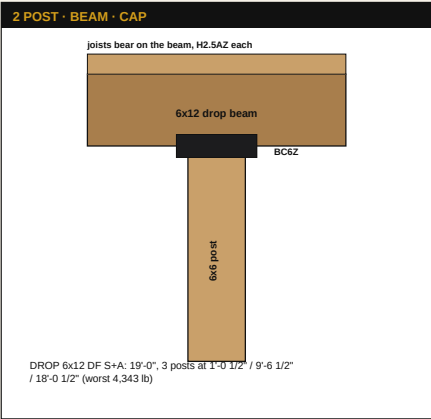
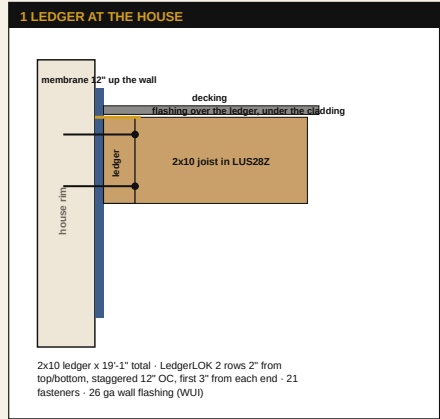


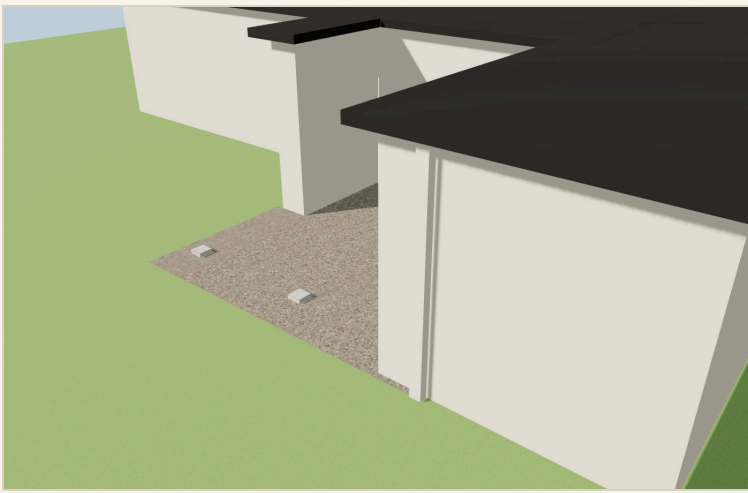
SECTION NOTES

Deck surface 2'-4" above grade
DROP 6x12 DF S+A: 19'-0", 3 posts at 1'-0 1/2" / 9'-6 1/2" / 18'-0 1/2" (worst 4,343 lb)
Posts 3 x 6x6 on DP-50/50
2x10 ledger x 19'-1" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 21 fasteners
G-Tape 2" every joist, PF joist, blocking and the ledger top · 4" on both rims and the beam(s)

Section left · elevation right · not to a common scale

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.





Step 1 · Footings — 3 Diamond Pier DP-50/50 driven-pin footings

T-400 · Step 1 · Footings

GOES IN

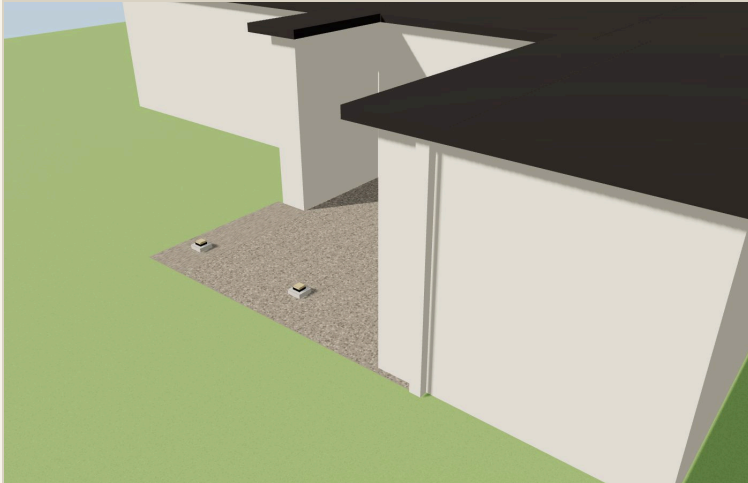
- (3) Diamond Pier DP-50/50 (head + 4 x 50" pins, plugs, caps)
- (3) Simpson ABA66Z adjustable post base 6x6 on the pier bolt

HOW

- Call 811 and locate before any hole or pin. Strip the existing deck first.
- Pull the beam lines from the house wall: DROP 6x12 DF S+A at 16'-1 3/16" from the reference wall, posts at 1'-0 1/2" / 9'-6 1/2" / 18'-0 1/2".
- Set 3 Diamond Pier heads level at grade on undisturbed soil and drive the 4 pins in each; caps on.

CHECK

- Footings on the layout within 1/2"; bearing on undisturbed soil
- Pins fully driven, heads level
- Photos before backfill



Step 2 · Posts — 6x6 #2 GC on ZMAX bases

T-401 · Step 2 · Posts

GOES IN

- (1) 6x6x8 #2 GC
- (3) Simpson ABA66Z adjustable post base 6x6 on the pier bolt

HOW

- 6x6 posts on the bases; cut so the beam top lands at the joist bottom (drop beam) or the joist top (flush beam): DROP 6x12 DF S+A posts 6" — field-measure each.
- Plumb both ways, brace, then the caps go on with the beams.

CHECK

- Posts plumb to a string; tops on the beam line
- No post in ground contact without a standoff

T-403 · Step 4 · Beams

GOES IN

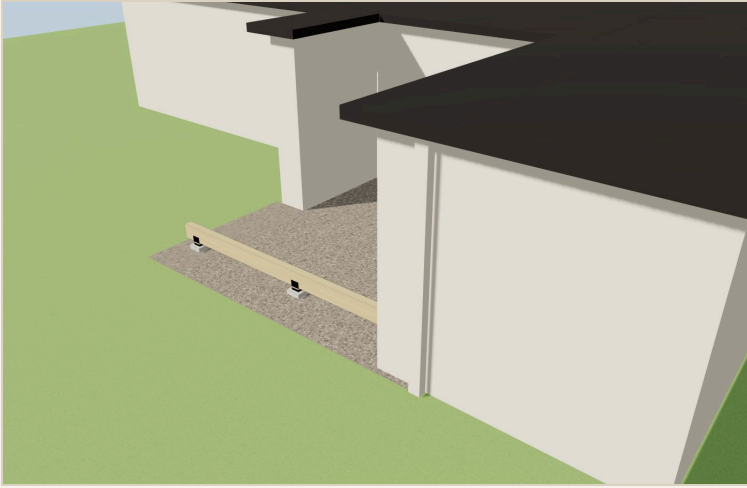
- (1) 6x12x20 Douglas Fir #2
- (1) 6x6x8 #2 GC
- (3) Simpson BC6Z post cap 6x post / 6x beam

HOW

- DROP 6x12 DF S+A: 19'-0" one piece, 3 posts @ 8'-6" max
- Crown up, strung straight, caps screwed off with the SDS screws in the box.

CHECK

- Beam top = joist bottom (drop) / joist top (flush)
- Splices land on a post; caps fully fastened



Step 4 · Beams — DROP 6x12 DF S+A (3 posts)

T-404 · Step 5 · Frame

GOES IN

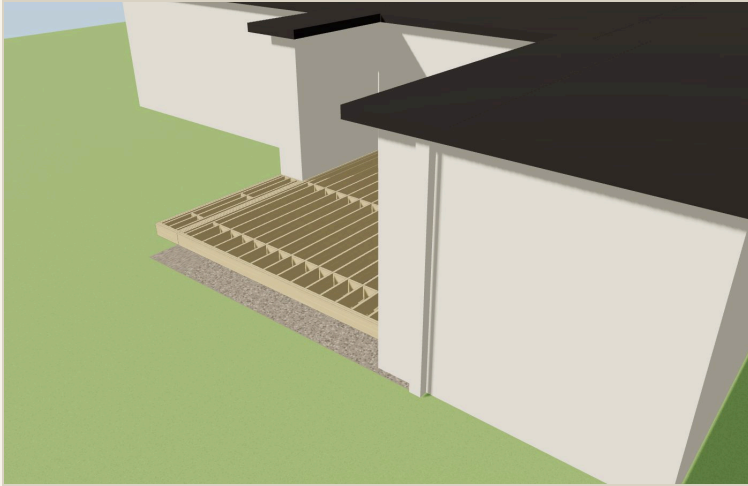
- (7) 2x10x10 #1 True Frame Joist SYP GC
- (4) 2x10x12 #1 True Frame Joist SYP GC
- (25) 2x10x20 #1 True Frame Joist SYP GC
- (42) Simpson LUS28Z face-mount hanger 2x8/2x10
- (8) Simpson HUCQ210-2-SDS concealed-flange double 2x10 hanger (SDS screws included)
- (1) Camo 9ga HDG hanger nail 3" 5 lb (~350)
- (1) Camo 9ga HDG hanger nail 1-1/2" 5 lb (~650)
- (1) FastenMaster LedgerLOK 3-5/8" 50 ct
- (25) Simpson H2.5AZ hurricane tie (joists to the drop beam, hidden)
- (1) Grace Vycor Deck Protector 12" x 75' (wall membrane behind ledger + side rims)
- (3) Aluminum L-flashing 10' (over ledger + side rims)

HOW

- 2x10 ledger x 19'-1" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 21 fasteners
- S: 1 joists @ 12" OC x 9'-9 1/2"; A: 16 joists @ 12" OC x 17'-11 1/2". Crown up, hangers at the ledger.
- Blocking one row over the beam, staggered.
- G-Tape 2" every joist, PF joist, blocking and the ledger top · 4" on both rims and the beam(s)

CHECK

- Frame square: diagonals within 1/8"
- Every hanger fully nailed; ledger fasteners per the schedule
- Tape continuous; photos of the ledger, hangers and footings before decking



Step 5 · Frame — 2x10 joists at 12", rims, ledger, blocking over the beam

T-405 · Step 6 · Decking

GOES IN

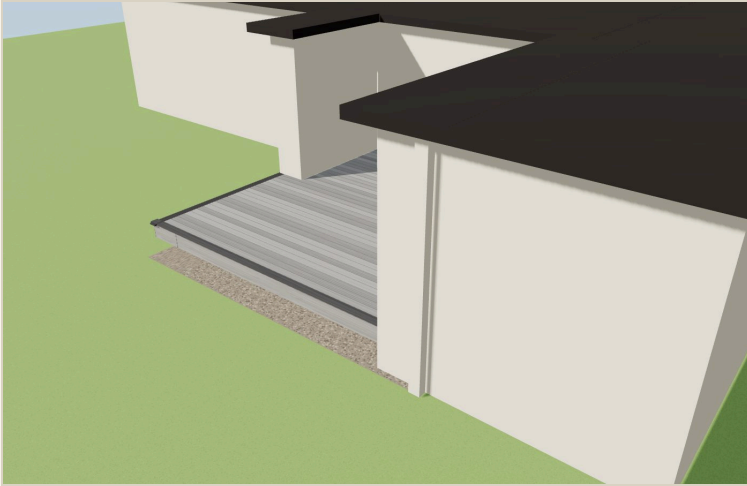
- (38) TimberTech Vintage Coastline 1x6x20 Square Edge
- (1) TimberTech Vintage Dark Hickory 1x6x16 Square Edge — borders
- (1) TimberTech Vintage Dark Hickory 1x6x20 Square Edge — borders
- (7) TimberTech Vintage Fascia Coastline 0.5x11.75x12
- (5) Cortex for TimberTech PVC, Coastline plugs (350 ct / 100 SF)
- (1) Cortex for TimberTech PVC, Dark Hickory plugs — borders + dividers

HOW

- Cortex face screws with color-matched plugs: 2 per board at every joist and at both ends of every run, color-matched plugs (1532 field + 50 border/divider)
- Borders first, then the field.
- Rows per panel: panel 1 (0"-19'-1") 38 rows · 1 ripped pieces · no breakers. Borders and breakers first, then the field rows from the outer edge toward the house.

CHECK

- 3/16" gaps held
- No cut end shows at the outer edge
- Color: every board from the same run



Step 6 · Decking — TimberTech Vintage Coastline, Dark Hickory border, hidden fasteners

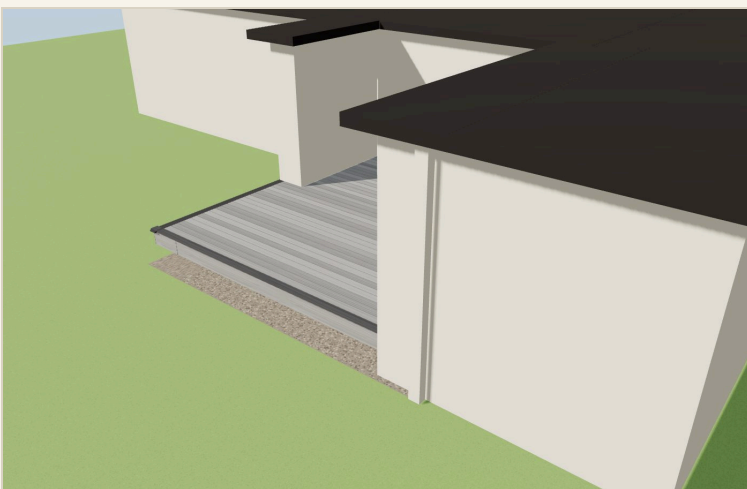
T-406 · Step 7 · Rail

GOES IN

HOW

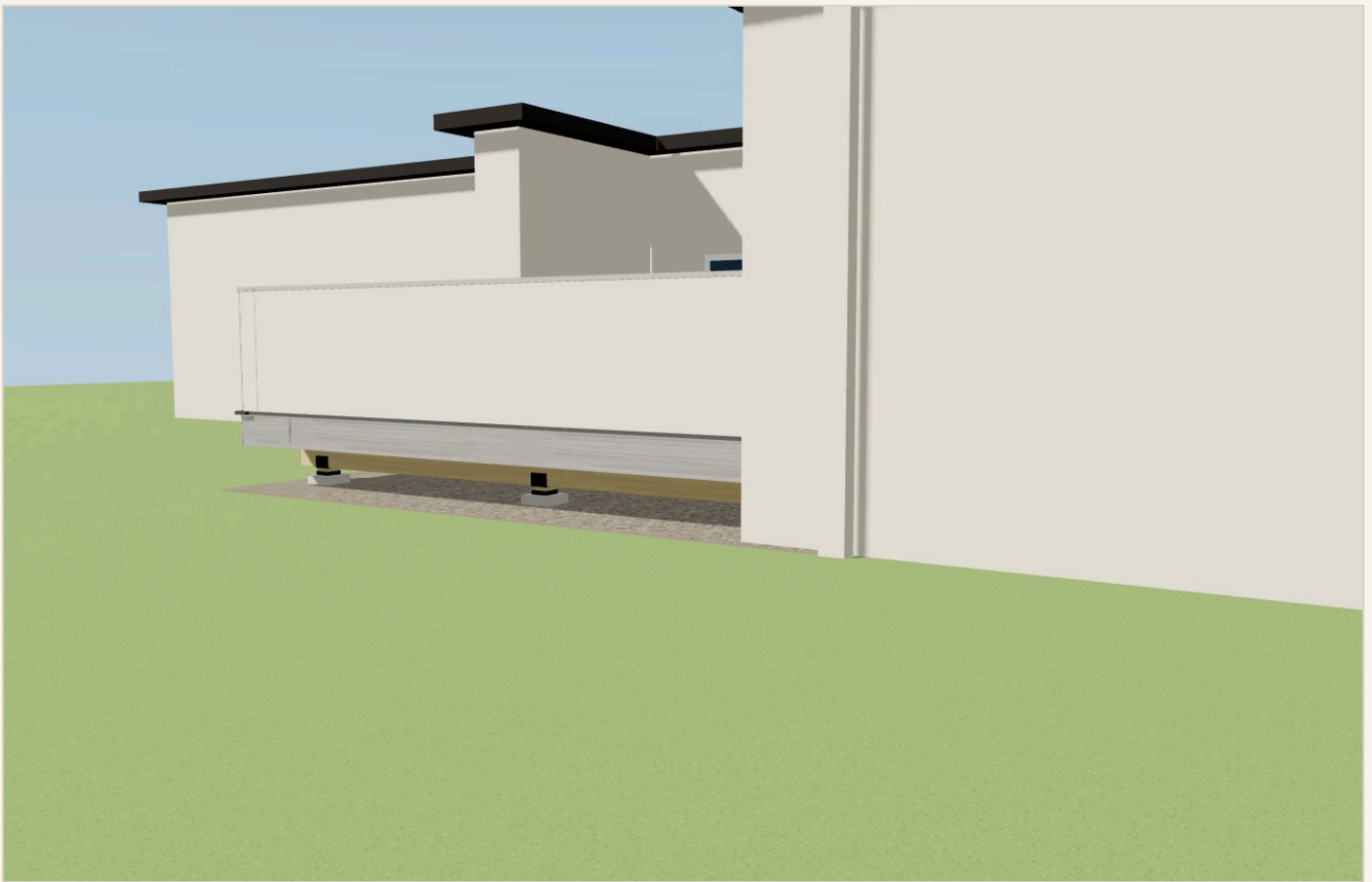
CHECK

- —
- Cables tensioned / balusters under 4" clear
- End of construction



Step 7 · Rail — existing stucco parapet stays

3D MODEL



The interactive 3D model lives at the online version of this document.

GS Exterior Experts · Deck Division

Prices: rate card · GC at cost · 42.5% GM · financed ÷ 0.93 unless the job says otherwise. Internal — not for the homeowner.